

Autonomous NAT Traversal

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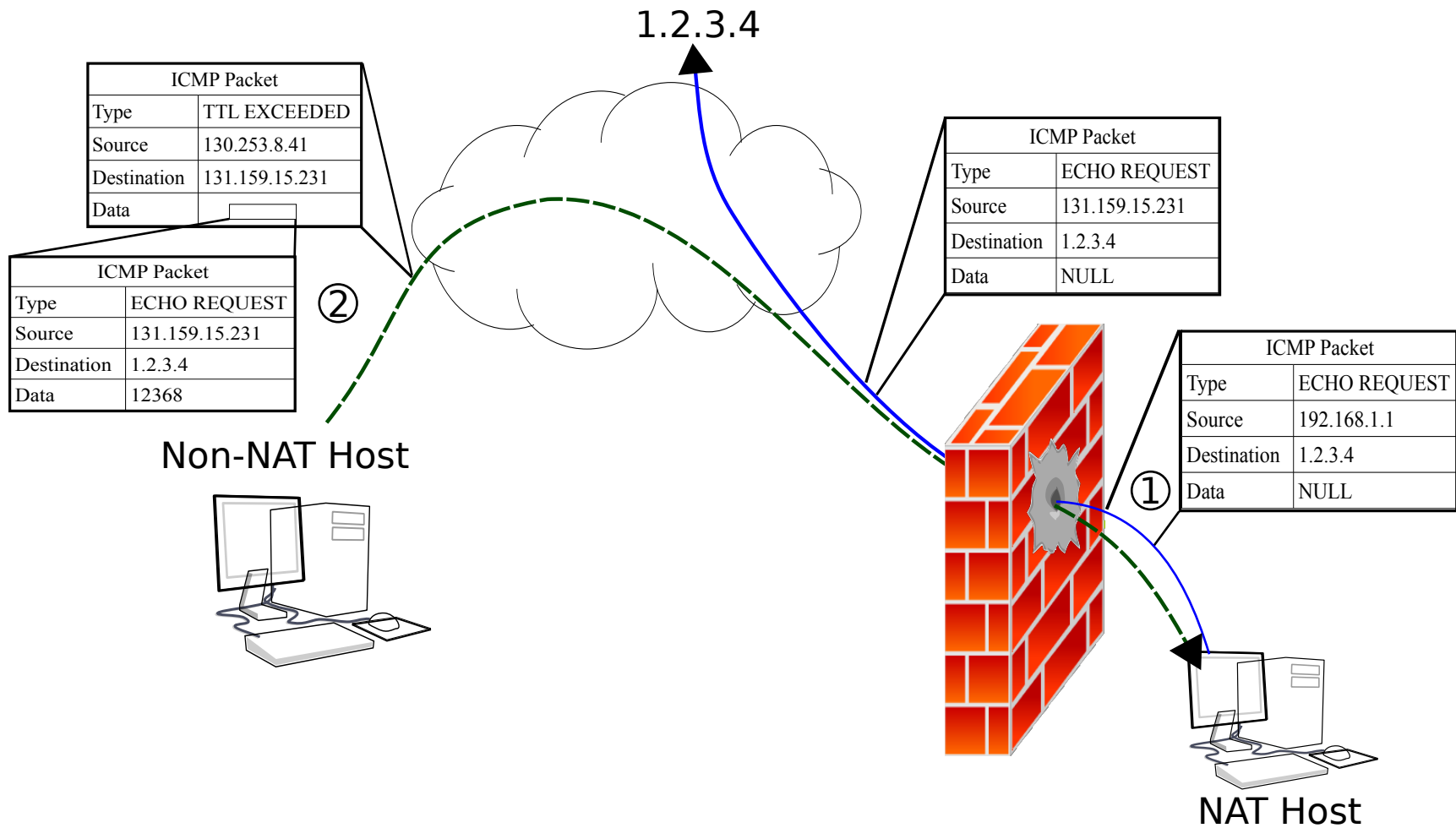
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Motivation

Enables initiation of connections to hosts behind NAT without involving a third party at the time.

- + Simpler to implement
- + Efficient, completely distributed method
- + Third party can not observe connections
- + Works well for UDP and TCP
 - Does not work as often as techniques involving 3rd party

Approach



Using UDP instead of ICMP ECHO REQUEST

- + No RAW sockets required for sending periodic requests
- + Might help punch hole
 - Slightly bigger messages
 - Smaller response payload (32 bits only)
 - May fail if NAT remaps ports

⇒ More experiments needed!

Experimental Setup

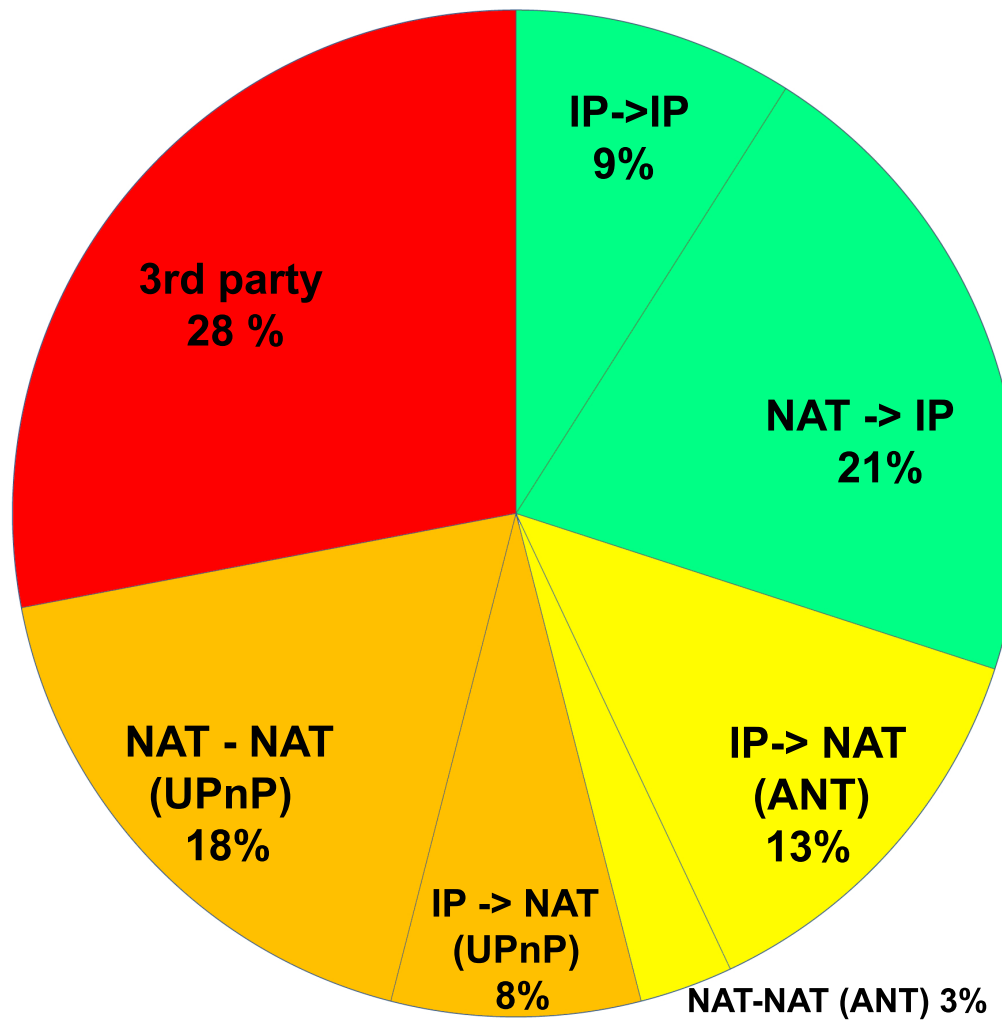
Internet community downloads NAT Tester client which:

- Attempts to transmit protocol messages to TUM server(s)
- Checks if protocol messages are received from TUM server(s)
- Requests user to enter NAT model information
- Submits results to TUM where they are added to public webpage

⇒ Large data set, somewhat biased selection of users

fsnsg

Experimental Results



Implementations

We implemented the proposed method three times:

NatTester Extensible framework for experiments

pwnat Stand-alone application for users

GNUnet Re-usable components for developers of P2P networks

All implementations are available as free software.

RTFL

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